

Complete Ordered Command Reference: RTL-to-GDSII ASIC Flow

Purpose

This document contains ONLY the commands used throughout the complete RTL-to-GDSII ASIC implementation flow in chronological order.

Each command includes:

- Purpose
- When to use it
- Expected result
- Important notes/warnings

The flow covers:

1. RTL Development
2. Simulation
3. Cadence Environment Setup
4. Xcelium/NCSim
5. Genus Synthesis
6. Innovus Physical Design
7. GDSII Generation

PHASE 1: LINUX + CADENCE ENVIRONMENT SETUP

Open TC Shell

```
tcsh
```

Purpose

Switches terminal into TC shell required by many Cadence setups.

Important Note

Some Cadence environment scripts may fail in normal bash shell.

Load Cadence Environment

```
source cshrc_new
```

Purpose

Loads:

- Cadence paths
- licenses
- environment variables
- EDA tool binaries

Expected Result

Cadence commands become available.

Verify Xcelium Installation

```
which xrun
```

Purpose

Checks whether Cadence simulation environment loaded correctly.

Expected Result

Displays path to xrun executable.

PHASE 2: PROJECT DIRECTORY SETUP

Create Project Directory

```
mkdir ALU_Project
```

Purpose

Creates working project directory.

Enter Project Directory

```
cd ALU_Project
```

Create RTL File

```
gedit ALU.v &
```

Purpose

Creates Verilog RTL file.

Create Testbench File

```
gedit tb_ALU.v &
```

Purpose

Creates Verilog testbench.

Verify Files

```
ls
```

Expected Result

```
ALU.v  
tb_ALU.v
```

PHASE 3: XCELIUM SIMULATION FLOW

Basic Terminal Simulation

```
xrun -R ALU.v tb_ALU.v
```

Purpose

Runs:

- compilation
- elaboration
- simulation

without GUI.

Important Note

This was the most stable simulation method.

GUI Simulation Attempt

```
xrun -gui ALU.v tb_ALU.v
```

Purpose

Launches SimVision waveform GUI.

Important Note

This caused runtime crashes in the lab environment.

Safer GUI Attempt

```
xrun -access +rwc -gui ALU.v tb_ALU.v
```

Purpose

Enables waveform read/write/connect access.

Important Note

Still unstable in current environment.

Clean Xcelium Files

```
rm -rf xcelium.d
```

```
rm -rf waves.shm
```

```
rm -rf xrun.*
```

Purpose

Removes:

- old simulation databases
 - waveform files
 - corrupted runtime data
-

PHASE 4: NCSIM FLOW (LEGACY FLOW ATTEMPT)

Compile Verilog

```
ncvlog ALU.v tb_ALU.v
```

Purpose

Compiles Verilog files.

Elaborate Design

```
ncelab tb_ALU
```

Purpose

Builds simulation hierarchy.

Run Simulation

```
ncsim tb_ALU
```

Purpose

Starts NCSim simulator.

Create Work Library

```
mkdir work
```

Purpose

Creates Cadence work library directory.

Create cds.lib

```
gedit cds.lib &
```

File Contents

```
DEFINE work ./work
```

Purpose

Defines Cadence library path.

Create hdl.var

```
gedit hdl.var &
```

File Contents

```
DEFINE WORK work
```

Purpose

Defines default working library.

PHASE 5: CADENCE GENUS SYNTHESIS FLOW

Launch Genus

```
genus &
```

Purpose

Starts Cadence synthesis environment.

Create Synthesis Script

```
gedit synth.tcl &
```

Synthesis TCL Script

```
read_hdl ALU.v  
  
elaborate ALU  
  
create_clock -period 10 clk  
  
synthesize -to_mapped  
  
report_area > area.rpt  
  
report_timing > timing.rpt  
  
report_power > power.rpt  
  
write_hdl > ALU_synth.v  
  
write_sdc > ALU.sdc
```

Run Synthesis Script

```
source synth.tcl
```

Purpose

Executes complete synthesis flow.

Verify Generated Files

```
ls
```

Expected Outputs

```
ALU_synth.v  
ALU.sdc  
area.rpt  
timing.rpt  
power.rpt
```

PHASE 6: PDK / TECHNOLOGY FILE SEARCH

Search LEF Files

```
find /opt/cadence -name "*.lef"
```

Purpose

Searches technology LEF and standard-cell LEF files.

Search Liberty Timing Files

```
find /opt/cadence -name "*.lib"
```

Purpose

Searches timing libraries.

Working Technology LEF

```
/opt/cadence/FOUNDRY/digital/45nm/dig/lef/gsclib045_tech.lef
```

Working Standard Cell LEF

```
/opt/cadence/FOUNDRY/digital/45nm/dig/lef/gsclib045_macro.lef
```

Working Timing Library

```
/opt/cadence/CONFRML211/share/cfm/lec/demo/rcv_intro/libs/gsclib045_v3.5/timing/  
slow.lib
```

PHASE 7: INNOVUS PHYSICAL DESIGN FLOW

Launch Innovus

```
innovus &
```

Purpose

Starts Cadence physical design environment.

Resume Suspended Innovus

```
fg
```

Purpose

Brings suspended Innovus process back to foreground.

Create Innovus TCL Script

```
gedit innovus.tcl &
```

Final Working Innovus Script

```
set init_lef_file {  
    /opt/cadence/FOUNDRY/digital/45nm/dig/lef/gsclib045_tech.lef  
    /opt/cadence/FOUNDRY/digital/45nm/dig/lef/gsclib045_macro.lef  
}  
  
set init_lib_file {  
    /opt/cadence/CONFRML211/share/cfm/lec/demo/rcv_intro/libs/gsclib045_v3.5/  
    timing/slow.lib  
}  
  
set init_verilog ALU_16_synth.v  
  
set init_top_cell ALU  
  
set init_sdc_file ALU.sdc  
  
init_design  
  
floorPlan -site CoreSite -r 1.0 0.7 20 20 20 20
```

```
place_design  
  
ccopt_design  
  
route_design  
  
report_area > innovus_area.rpt  
  
report_timing > innovus_timing.rpt  
  
saveDesign ALU_innovus.enc
```

Run Innovus Script

```
source innovus.tcl
```

Purpose

Executes:

- initialization
- floorplanning
- placement
- CTS
- routing

Verify Timing

```
report_timing
```

Purpose

Displays timing analysis.

Verify DRC

```
verify_drc
```

Purpose

Checks physical design rule violations.

Verify Connectivity

```
verifyConnectivity
```

Purpose

Checks:

- opens
 - shorts
 - disconnected pins
-

Save Design Database

```
saveDesign final_ALU.enc
```

Purpose

Saves Innovus design database.

Export Final Netlist

```
saveNetlist final_ALU.v
```

Purpose

Exports routed gate-level netlist.

Generate GDSII

```
streamOut final_ALU.gds -libName ALU
```

Purpose

Generates final GDSII layout file.

Expected Result

```
#####Streamout is finished!
```

PHASE 8: DOCUMENTATION + ORGANIZATION

Create Documentation Folders

```
mkdir documentation
```

```
mkdir screenshots
```

```
mkdir reports
```

```
mkdir logs
```

```
mkdir GDS
```

```
mkdir RTL
```

Move Reports

```
mv *.rpt reports/
```

Move Logs

```
mv *.log logs/
```

```
mv *.err logs/
```

Move RTL Files

```
mv ALU.v RTL/
```

```
mv tb_ALU.v RTL/
```

Move Final GDS Files

```
mv final_ALU* GDS/
```

IMPORTANT WARNINGS AND NOTES

Note 1

Cadence GUI crashes were tool/environment issues, NOT RTL issues.

Note 2

Successful compile + elaboration usually confirms RTL correctness.

Note 3

Innovus REQUIRES:

- LEF files
- LIB files
- technology setup

Genus synthesis alone is insufficient.

Note 4

Commands like:

```
floorPlan
```

must ONLY be executed inside:

```
innovus>
```

prompt.

Note 5

Incomplete timing constraints caused:

- unconstrained paths
- incomplete CTS

- timing optimization limitations
-

Note 6

Even with incomplete MMMC setup:

- placement
- routing
- GDS generation

still completed successfully.

Note 7

Multipliers significantly increase:

- area
 - routing complexity
 - power consumption
-

Note 8

The final backend flow was academic/physical-only style, not full signoff-quality implementation.

COMPLETE FLOW SUMMARY

```
tcsn
  ↓
source cshrc_new
  ↓
RTL creation
  ↓
Xcelium simulation
  ↓
Genus synthesis
  ↓
PDK setup
  ↓
```

Innovus floorplanning



Placement



CTS



Routing



DRC/connectivity



GDSII generation